Name : Aditi Bhoir

124A7004

B3

EXPERIMENT NO. 8
CONFIGURATION OF HYBRID ROUTING PROTOCOL (EIGRP)

AIM

To study and implement EIGRP routing protocol and analyze its features.

OBJECTIVES

- To configure EIGRP on multiple routers.
- To study composite metrics used by EIGRP.
- To analyze fast convergence behavior.
- To compare EIGRP with other routing protocols.
- To verify end-to-end connectivity.

THEORY

Enhanced Interior Gateway Routing Protocol is an advanced hybrid routing protocol developed by Cisco. It combines features of distance vector and link state protocols.

EIGRP uses DUAL algorithm and composite metrics such as bandwidth and delay to select best routes.

SOFTWARE / MATERIALS REQUIRED

- Cisco Packet Tracer 8.x
- Computer System

IMPORTANT COMMANDS / PROGRAM STEPS

```
router eigrp 600
network 10.0.0.0
network 172.16.0.0
no auto-summary
show ip eigrp neighbors
show ip eigrp topology
show ip eigrp traffic
show ip route eigrp
bandwidth 256        (interface mode)
delay 30             (interface mode)
ping 30.0.0.2
```

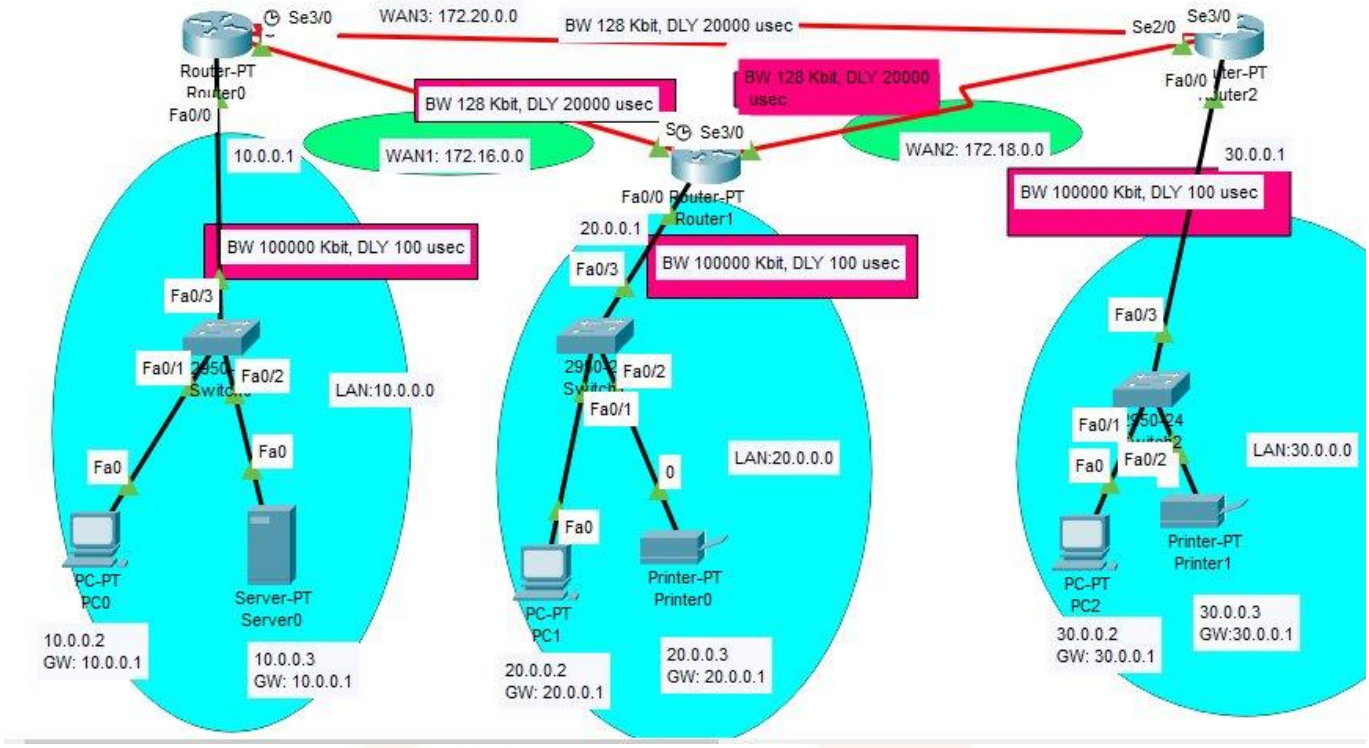
PROCEDURE

1. Configure IP addresses on routers and PCs.
2. Enable EIGRP with the same autonomous system number on all routers.
3. Advertise connected networks and disable auto-summary.
4. Verify EIGRP neighbors and topology table.
5. Check routing table and test connectivity.
6. Change interface bandwidth and delay and observe the change in metric and path selection.
7. Activate the third WAN link and observe re-convergence.

OBSERVATION / COMPARISON TABLES:

Parameter	**RIP**	**OSPF**	**EIGRP**
Full Form	Routing Information Protocol	Open Shortest Path First	Enhanced Interior Gateway Routing Protocol
Type	Distance Vector	Link State	Advanced Distance Vector (often called Hybrid)
Algorithm	Bellman-Ford	Dijkstra (SPF)	DUAL
Metric Used	Hop count	Cost (based on bandwidth)	Composite metric, mainly bandwidth and delay
Maximum Hop Count	15 hops	No fixed hop-count limit	No fixed hop-count limit like RIP
Convergence Speed	Slow	Fast	Very fast
Routing Updates	Periodic updates	Link-state advertisements (LSAs) and triggered updates	Partial, bounded updates
Network Knowledge	Routing information from neighboring routers	Complete topology database within an area	Neighbor and topology tables
Scalability	Small networks	Medium to large networks	Medium to large networks
Classless Routing	RIPv2 supports VLSM and CIDR	Supports VLSM and CIDR	Supports VLSM and CIDR
Administrative Distance (Cisco)	120	110	90 (internal EIGRP)
Route Selection	Lowest hop count	Lowest path cost	Lowest feasible composite metric
Configuration Complexity	Simple	Moderate	Moderate
Cisco Route Code	`R`	`O`	`D`

RESULT:


Se3/0
WAN3: 172.20.0.0
BW 128 Kbit, DLY 20000 usec
Se2/0
Se3/0
Router-PT
Fa0/0
BW 128 Kbit, DLY 20000 usec
BW 128 Kbit, DLY 20000 usec
Fa0/0
10.0.0.1
WAN1: 172.16.0.0
Se3/0
WAN2: 172.18.0.0
30.0.0.1
Fa0/0 Router-PT
Router1
BW 100000 Kbit, DLY 100 usec
20.0.0.1
BW 100000 Kbit, DLY 100 usec
Fa0/3
Fa0/3
Fa0/1
Fa0/2
Fa0/2
LAN:10.0.0.0
Fa0/1
Fa0
Fa0
LAN:20.0.0.0
0
Fa0
Fa0/1
Fa0/2
LAN:30.0.0.0
PC-PT
PC0
Server-PT
Server0
Printer-PT
Printer0
PC-PT
PC1
Printer-PT
Printer1
PC-PT
PC2
10.0.0.2
GW: 10.0.0.1
10.0.0.3
GW: 10.0.0.1
20.0.0.2
GW: 20.0.0.1
20.0.0.3
GW: 20.0.0.1
30.0.0.2
GW: 30.0.0.1
30.0.0.3
GW:30.0.0.1

2. EIGRP Protocol Configuration on Routers

Router0

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in fa 0/0
Router(config-if)#ip address 10.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#in se 2/0
Router(config-if)#ip address 172.16.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
Router(config-if)#in se 3/0
Router(config-if)#ip addr 172.20.01 255.255.0.0
                          ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 172.20.01 255.255.0.0
                             ^
% Invalid input detected at '^' marker.

Router(config-if)#ip address 172.20.0.1 255.255.0.0
Router(config-if)#
```

Router1

```
Router>en
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in fa 0/0
Router(config-if)#ip address 20.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#in se 3/0
Router(config-if)#ip address 172.18.0.1 255.255.0.0
Router(config-if)#clock rate 64000
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#

Router(config-if)#in se 2/0
Router(config-if)#ip address 172.16.0.2 255.255.0.0
```

Router2

```
Router>en
Router#
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in fa 0/0
Router(config-if)#ip address 30.0.0.1 255.0.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#in se 2/0
Router(config-if)#ip address 172.18.0.2 255.255.0.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

Router(config-if)#in se 3/
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed statno shut
Router(config-if)#in se 3/0
Router(config-if)#ip address 172.20.0.2 255.255.0.0
Router(config-if)#
```

2. Routing table Before EIGRP Routing

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    20.0.0.0/8 is directly connected, FastEthernet0/0
C    172.16.0.0/16 is directly connected, Serial2/0
C    172.18.0.0/16 is directly connected, Serial3/0

Router#
```

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
C    172.16.0.0/16 is directly connected, Serial2/0

Router#
```

```
Router>en
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    30.0.0.0/8 is directly connected, FastEthernet0/0
C    172.18.0.0/16 is directly connected, Serial2/0

Router#
```

3. EIGRP Routing Configuration

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 600
Router(config-router)#network 30.0.0.0
Router(config-router)#network 172.18.0.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.18.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 172.20.0.0
Router(config-router)#
```

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 600
Router(config-router)#network 20.0.0.0
Router(config-router)#network 172.16.0.0
Router(config-router)#
%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.16.0.1 (Serial2/0) is up: new adjacency

Router(config-router)#network 172.18.0.0
Router(config-router)#
```

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router eigrp 600
Router(config-router)#network 10.0.0.0
Router(config-router)#network 172.16.0.0
Router(config-router)#network 172.20.0.0
Router(config-router)#
```

3. Interface Bandwidth Verification and Customization

Router0

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in se 2/0
Router(config-if)#bandwidth 256
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.16.0.2 (Serial2/0) is down: interface down

%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.16.0.2 (Serial2/0) is up: new adjacency

Router(config-if)#in se 3/0
Router(config-if)#bandwidth 256
Router(config-if)#
```

```
Router#sh int se 3/0
Serial3/0 is administratively down, line protocol is down (disabled)
  Hardware is HD64570
  Internet address is 172.20.0.1/16
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 2 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=down  DSR=down  DTR=down  RTS=down  CTS=down
```

```
Router#sh int se 2/0
Serial2/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 172.16.0.1/16
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 70 bits/sec, 0 packets/sec
  5 minute output rate 92 bits/sec, 0 packets/sec
     52 packets input, 3076 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     62 packets output, 3649 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=up  DSR=up  DTR=up  RTS=up  CTS=up
```

```
Router#sh int fa 0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 00e0.8f39.6076 (bia 00e0.8f39.6076)
  Internet address is 10.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 88 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     55 packets output, 3300 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
```

Router1

```
Router#sh int se 3/0
Serial3/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 172.18.0.1/16
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 102 bits/sec, 0 packets/sec
  5 minute output rate 104 bits/sec, 0 packets/sec
     72 packets input, 4316 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     80 packets output, 4729 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=up  DSR=up  DTR=up  RTS=up  CTS=up
Router#sh int se 2/0
Serial2/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 172.16.0.2/16
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 107 bits/sec, 0 packets/sec
  5 minute output rate 105 bits/sec, 0 packets/sec
     105 packets input, 6272 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     86 packets output, 5058 bytes, 0 underruns
     0 output errors, 0 collisions, 2 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=up  DSR=up  DTR=up  RTS=up  CTS=up
```

```
Router#sh int fa 0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0006.2a79.1c4b (bia 0006.2a79.1c4b)
  Internet address is 20.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 102 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     77 packets output, 4620 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
```

```
Router(config)#in se 3/0
Router(config-if)#bandwidth
% Incomplete command.
Router(config-if)#bandwidth 256
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.18.0.2 (Serial3/0) is down: interface down

%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.18.0.2 (Serial3/0) is up: new adjacency

Router(config-if)#delay 30
Router(config-if)#
%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.18.0.2 (Serial3/0) is down: interface down

%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.18.0.2 (Serial3/0) is up: new adjacency

Router(config-if)#
```

Router2

```
Router#sh int se 3/0
Serial3/0 is administratively down, line protocol is down (disabled)
  Hardware is HD64570
  Internet address is 172.20.0.2/16
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 2 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=down  DSR=down  DTR=down  RTS=down  CTS=down
```

```
Router#sh int se 2/0
Serial2/0 is up, line protocol is up (connected)
  Hardware is HD64570
  Internet address is 172.18.0.2/16
  MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation HDLC, loopback not set, keepalive set (10 sec)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: weighted fair
  Output queue: 0/1000/64/0 (size/max total/threshold/drops)
     Conversations  0/0/256 (active/max active/max total)
     Reserved Conversations 0/0 (allocated/max allocated)
     Available Bandwidth 1158 kilobits/sec
  5 minute input rate 110 bits/sec, 0 packets/sec
  5 minute output rate 106 bits/sec, 0 packets/sec
     122 packets input, 7455 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     103 packets output, 6078 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 output buffer failures, 0 output buffers swapped out
     0 carrier transitions
     DCD=up  DSR=up  DTR=up  RTS=up  CTS=up
```

```
Router#sh int fa 0/0
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 00d0.58d2.e714 (bia 00d0.58d2.e714)
  Internet address is 30.0.0.1/8
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 104 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 input packets with dribble condition detected
     95 packets output, 5700 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 babbles, 0 late collision, 0 deferred
     0 lost carrier, 0 no carrier
     0 output buffer failures, 0 output buffers swapped out
```

4. EIGRP Neighbor Relationships, Topology Table and Traffic

Router0

```
Router#sh ip eigrp traffic
IP-EIGRP Traffic Statistics for process 600
  Hellos sent/received: 294/135
  Updates sent/received: 12/9
  Queries sent/received: 0/4
  Replies sent/received:  4/0
  Acks sent/received:  13/16
  Input queue high water mark 1, 0 drops
  SIA-Queries sent/received: 0/0
  SIA-Replies sent/received: 0/0

Router#
```

```
Router#sh ip eigrp topology
IP-EIGRP Topology Table for AS 600/ID(172.16.0.1)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/8, 1 successors, FD is 28160
         via Connected, FastEthernet0/0
P 20.0.0.0/8, 1 successors, FD is 10514432
         via 172.16.0.2 (10514432/28160), Serial2/0
P 30.0.0.0/8, 1 successors, FD is 10522112
         via 172.16.0.2 (10522112/10010112), Serial2/0
P 172.16.0.0/16, 1 successors, FD is 10511872
         via Connected, Serial2/0
P 172.18.0.0/16, 1 successors, FD is 10519552
         via 172.16.0.2 (10519552/10007552), Serial2/0
Router#
```

```
Router#sh ip eigrp neighbor
IP-EIGRP neighbors for process 600
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)       Cnt  Num
0   172.16.0.2      Se2/0          15   00:05:29  40     1000  0   22

Router#
```

```
Router#sh ip route eigrp
D    20.0.0.0/8 [90/10514432] via 172.16.0.2, 00:05:07, Serial2/0
D    30.0.0.0/8 [90/10522112] via 172.16.0.2, 00:02:18, Serial2/0
D    172.18.0.0/16 [90/10519552] via 172.16.0.2, 00:02:19, Serial2/0

Router#
```

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
D    20.0.0.0/8 [90/10514432] via 172.16.0.2, 00:04:50, Serial2/0
D    30.0.0.0/8 [90/10522112] via 172.16.0.2, 00:02:01, Serial2/0
C    172.16.0.0/16 is directly connected, Serial2/0
D    172.18.0.0/16 [90/10519552] via 172.16.0.2, 00:02:02, Serial2/0

Router#
```

Router1

```
Router#sh ip eigrp traffic
IP-EIGRP Traffic Statistics for process 600
  Hellos sent/received: 441/281
  Updates sent/received: 18/18
  Queries sent/received: 4/0
  Replies sent/received:  0/4
  Acks sent/received:  22/19
  Input queue high water mark 1, 0 drops
  SIA-Queries sent/received: 0/0
  SIA-Replies sent/received: 0/0

Router#
```

```
Router#sh ip eigrp topology
IP-EIGRP Topology Table for AS 600/ID(172.18.0.1)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/8, 1 successors, FD is 2172416
         via 172.16.0.1 (2172416/28160), Serial2/0
P 20.0.0.0/8, 1 successors, FD is 28160
         via Connected, FastEthernet0/0
P 30.0.0.0/8, 1 successors, FD is 10010112
         via 172.18.0.2 (10010112/28160), Serial3/0
P 172.16.0.0/16, 1 successors, FD is 2169856
         via Connected, Serial2/0
P 172.18.0.0/16, 1 successors, FD is 10007552
         via Connected, Serial3/0
Router#
```

```
Router#sh ip eigrp neighbor
IP-EIGRP neighbors for process 600
H   Address         Interface       Hold Uptime    SRTT   RTO   Q   Seq
                                    (sec)          (ms)       Cnt  Num
0   172.16.0.1      Se2/0           12   00:10:50  40     1000  0   26
1   172.18.0.2      Se3/0           13   00:03:43  40     1000  0   9

Router#
```

```
Router#sh ip route eigrp
D    10.0.0.0/8 [90/2172416] via 172.16.0.1, 00:10:38, Serial2/0
D    30.0.0.0/8 [90/10010112] via 172.18.0.2, 00:03:31, Serial3/0

Router#
```

```
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

D    10.0.0.0/8 [90/2172416] via 172.16.0.1, 00:10:30, Serial2/0
C    20.0.0.0/8 is directly connected, FastEthernet0/0
D    30.0.0.0/8 [90/10010112] via 172.18.0.2, 00:03:23, Serial3/0
C    172.16.0.0/16 is directly connected, Serial2/0
C    172.18.0.0/16 is directly connected, Serial3/0

Router#
```

Router2

```
Router#sh ip eigrp traffic
IP-EIGRP Traffic Statistics for process 600
  Hellos sent/received: 291/148
  Updates sent/received: 6/8
  Queries sent/received: 0/0
  Replies sent/received:  0/0
  Acks sent/received:  8/6
  Input queue high water mark 1, 0 drops
  SIA-Queries sent/received: 0/0
  SIA-Replies sent/received: 0/0

Router#
```

```
Router#sh ip eigrp topology
IP-EIGRP Topology Table for AS 600/ID(172.18.0.2)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/8, 1 successors, FD is 2684416
         via 172.18.0.1 (2684416/2172416), Serial2/0
P 20.0.0.0/8, 1 successors, FD is 2172416
         via 172.18.0.1 (2172416/28160), Serial2/0
P 30.0.0.0/8, 1 successors, FD is 28160
         via Connected, FastEthernet0/0
P 172.16.0.0/16, 1 successors, FD is 2681856
         via 172.18.0.1 (2681856/2169856), Serial2/0
P 172.18.0.0/16, 1 successors, FD is 2169856
         via Connected, Serial2/0
```

```
Router#sh ip eigrp neighbor
IP-EIGRP neighbors for process 600
H   Address         Interface      Hold Uptime    SRTT   RTO   Q   Seq
                                   (sec)          (ms)        Cnt  Num
0   172.18.0.1      Se2/0          12   00:10:51  40     1000  0   23
```

```
Router#sh ip route eigrp
D    10.0.0.0/8 [90/2684416] via 172.18.0.1, 00:10:44, Serial2/0
D    20.0.0.0/8 [90/2172416] via 172.18.0.1, 00:10:44, Serial2/0
D    172.16.0.0/16 [90/2681856] via 172.18.0.1, 00:10:44, Serial2/0
```

```
Router>en
Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

D    10.0.0.0/8 [90/2684416] via 172.18.0.1, 00:10:38, Serial2/0
D    20.0.0.0/8 [90/2172416] via 172.18.0.1, 00:10:38, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
D    172.16.0.0/16 [90/2681856] via 172.18.0.1, 00:10:38, Serial2/0
C    172.18.0.0/16 is directly connected, Serial2/0
```

5. Updated Routing Table Observations

After EIGRP Routing

Router0

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#in se 3/0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.20.0.1 (Serial3/0) is up: new adjacency

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

D    10.0.0.0/8 [90/2172416] via 172.20.0.1, 00:01:00, Serial3/0
D    20.0.0.0/8 [90/2172416] via 172.18.0.1, 00:13:22, Serial2/0
C    30.0.0.0/8 is directly connected, FastEthernet0/0
D    172.16.0.0/16 [90/2681856] via 172.18.0.1, 00:13:22, Serial2/0
C    172.18.0.0/16 is directly connected, Serial2/0
C    172.20.0.0/16 is directly connected, Serial3/0
```

Router2

```
Router(config)#in se 3/0
Router(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
Router(config-if)#
%LINK-5-CHANGED: Interface Serial3/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial3/0, changed state to up

%DUAL-5-NBRCHANGE: IP-EIGRP 600: Neighbor 172.20.0.2 (Serial3/0) is up: new adjacency

Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#sh ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
D    20.0.0.0/8 [90/10514432] via 172.16.0.2, 00:09:35, Serial2/0
D    30.0.0.0/8 [90/10514432] via 172.20.0.2, 00:00:32, Serial3/0
C    172.16.0.0/16 is directly connected, Serial2/0
D    172.18.0.0/16 [90/10519552] via 172.16.0.2, 00:06:47, Serial2/0
C    172.20.0.0/16 is directly connected, Serial3/0
```

Ping response

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.3

Pinging 10.0.0.3 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.3: bytes=32 time=20ms TTL=126
Reply from 10.0.0.3: bytes=32 time=15ms TTL=126
Reply from 10.0.0.3: bytes=32 time=14ms TTL=126

Ping statistics for 10.0.0.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 20ms, Average = 16ms

C:\>
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 30.0.0.2

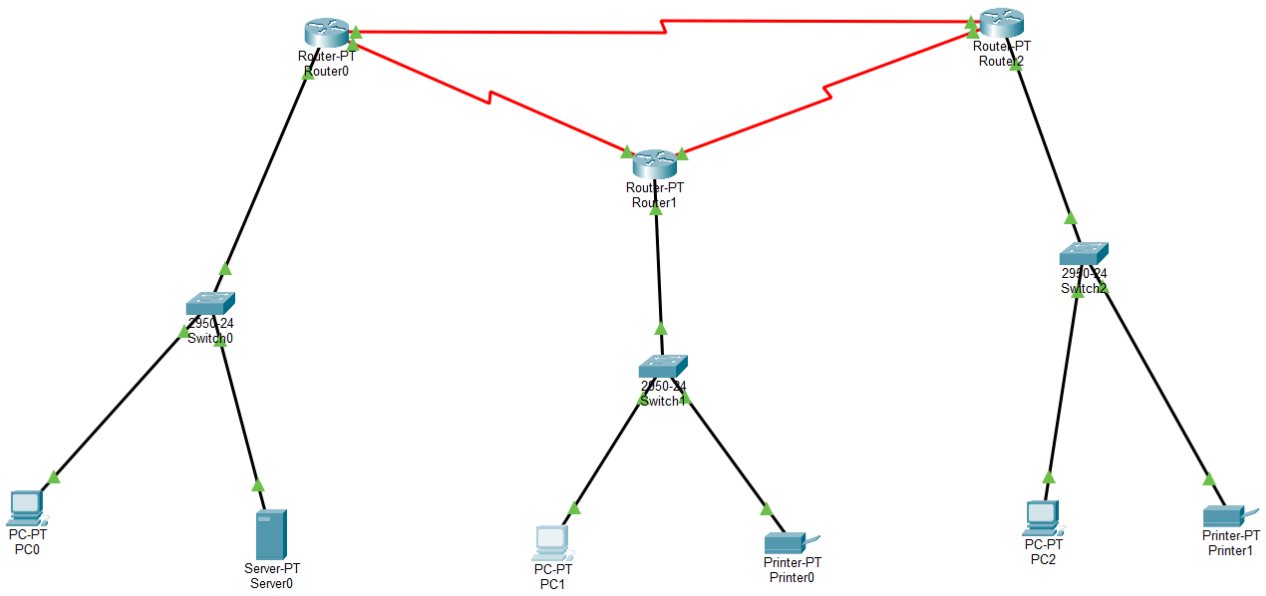
Pinging 30.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 30.0.0.2: bytes=32 time=8ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126
Reply from 30.0.0.2: bytes=32 time=6ms TTL=126

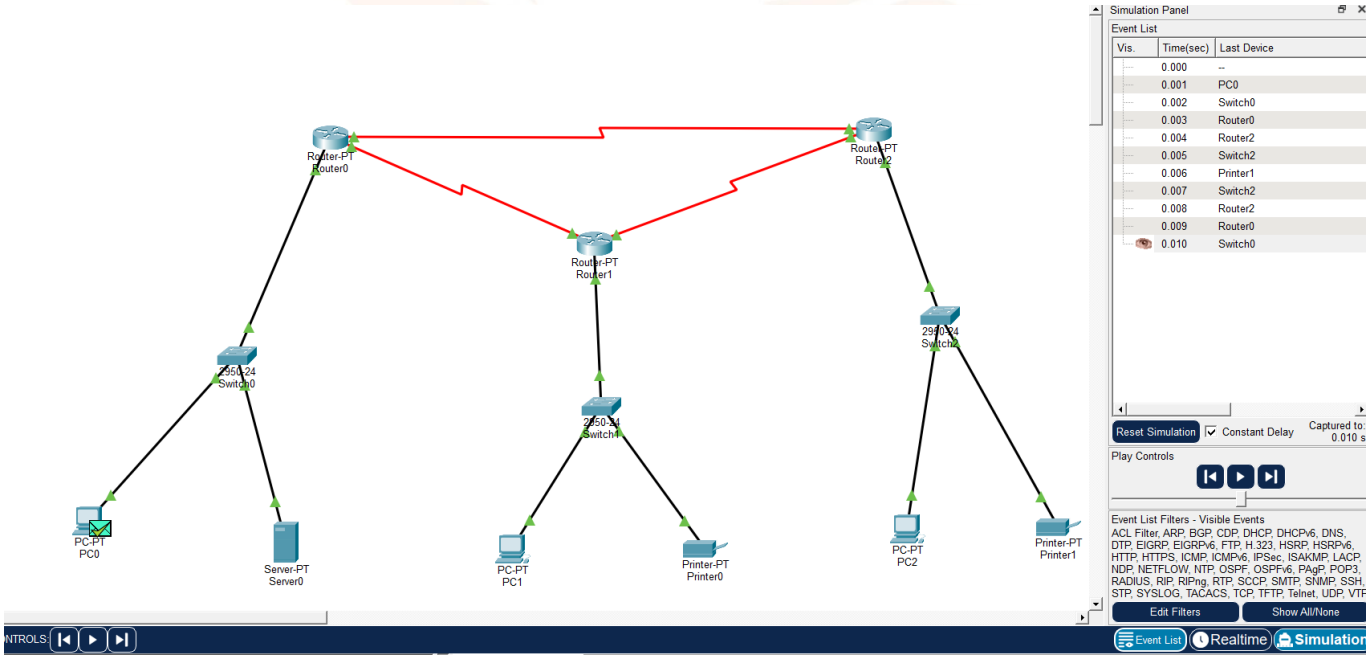
Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 8ms, Average = 6ms

C:\>
```

Sequence Flow:



PDU



CONCLUSION:

This experiment demonstrated the configuration of EIGRP (AS 600) on three routers using Cisco Packet Tracer. All routers formed neighbor adjacencies and exchanged routes using the DUAL algorithm, building complete topology tables with successor routes and feasible distances. Changing the interface bandwidth and delay altered the composite metric and directly influenced path selection. Activating the third WAN link triggered quick re-convergence to a lower-metric path without any manual intervention. Successful ping tests confirmed end-to-end connectivity, showing that EIGRP is fast-converging and efficient.